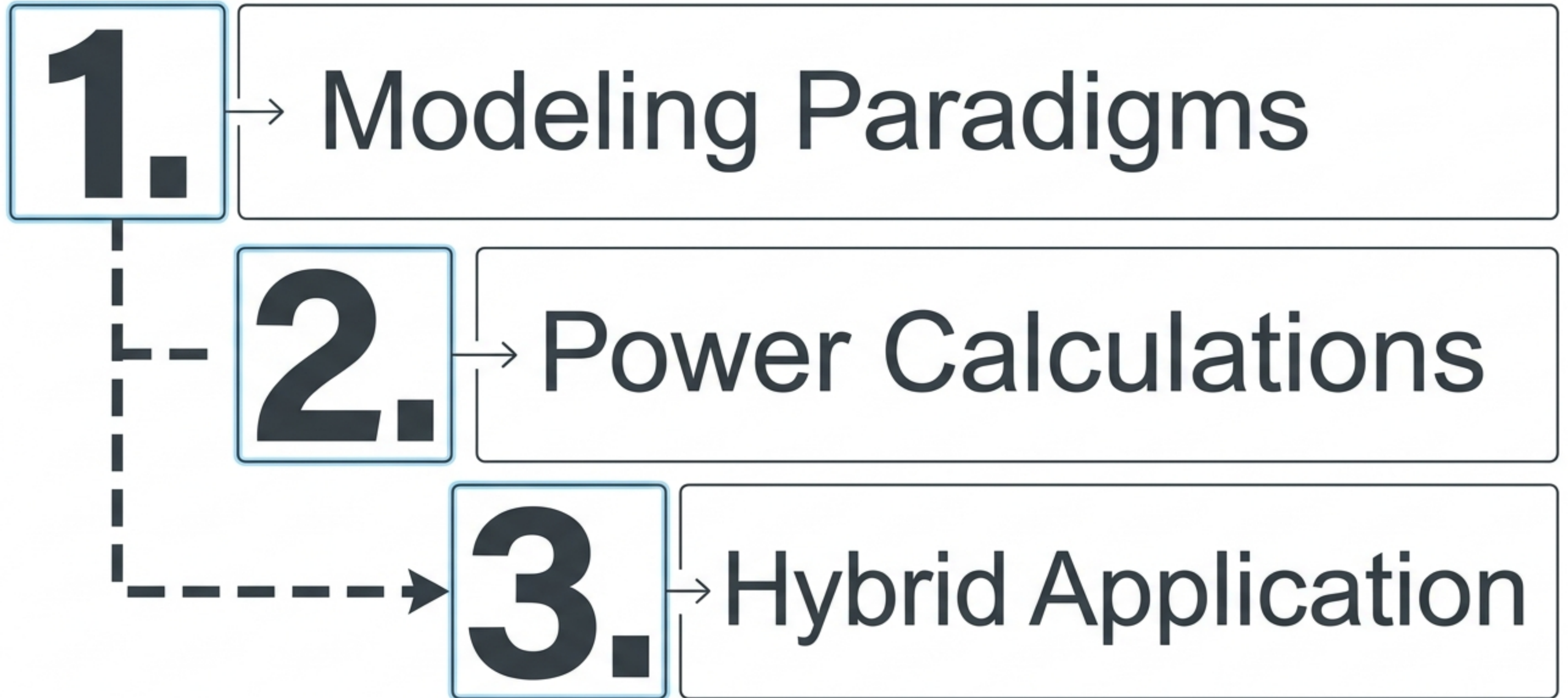


MODULE 2, LESSON 1

AI 101 FOR MECHANICAL ENGINEERS

PHYSICS, DATA AND HYBRID MODELS

Lesson Outline



1. Modeling Paradigms

Distinguishing physical, data-driven,
and hybrid engineering models.

Core Modeling Paradigms

Physics-based



Utilizes governing physical relationships and stated assumptions.

Data-driven



Infers relationships directly from observed input-output data.

Hybrid



Combines physical structural baselines with learned components.

Selection Parameters



Evaluate available parameter knowledge and measurement quality.



Define the required operational domain and range.



Assess the ultimate engineering purpose of the model.



Lower error rates do not validate flawed inputs.

Defining Simulation



Simulation executes a mathematical model computationally.

Execution requires defined initial and boundary conditions.

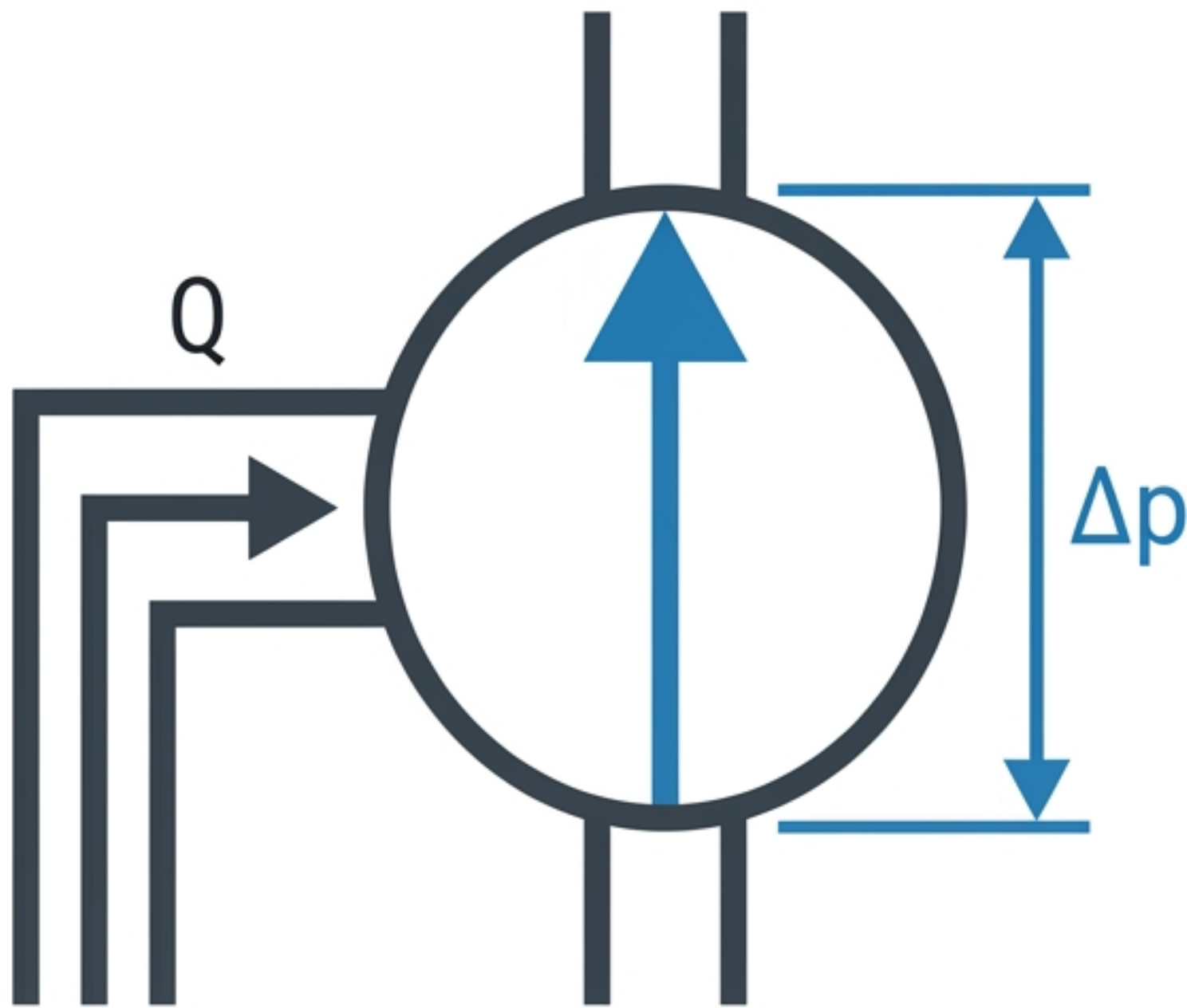
Simulation analyzes transient or steady-state system behavior.

Example application: Modeling heat-transfer transients over time.

2. Power Calculations

Formulating mathematical baselines
for fluid systems.

Calculating Hydraulic Power



$$P_h = \Delta p \times Q$$

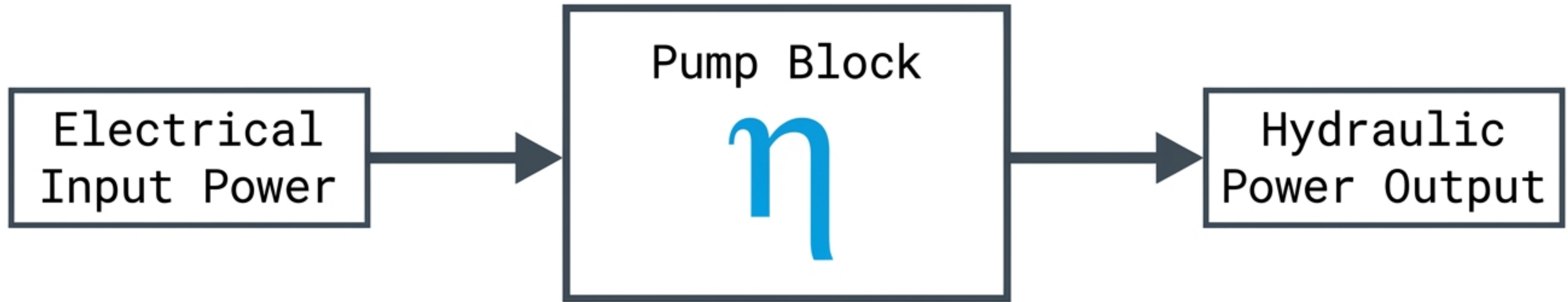
P_h : Hydraulic power (W)

Δp : Pressure rise (Pa)

Q : Volumetric flow (m^3/s)

Unit note: 1 Pa = 1 N/m²

Determining Input Power



$$P_{in} = P_h / \eta$$

P_{in} : Input shaft or electrical power (W).

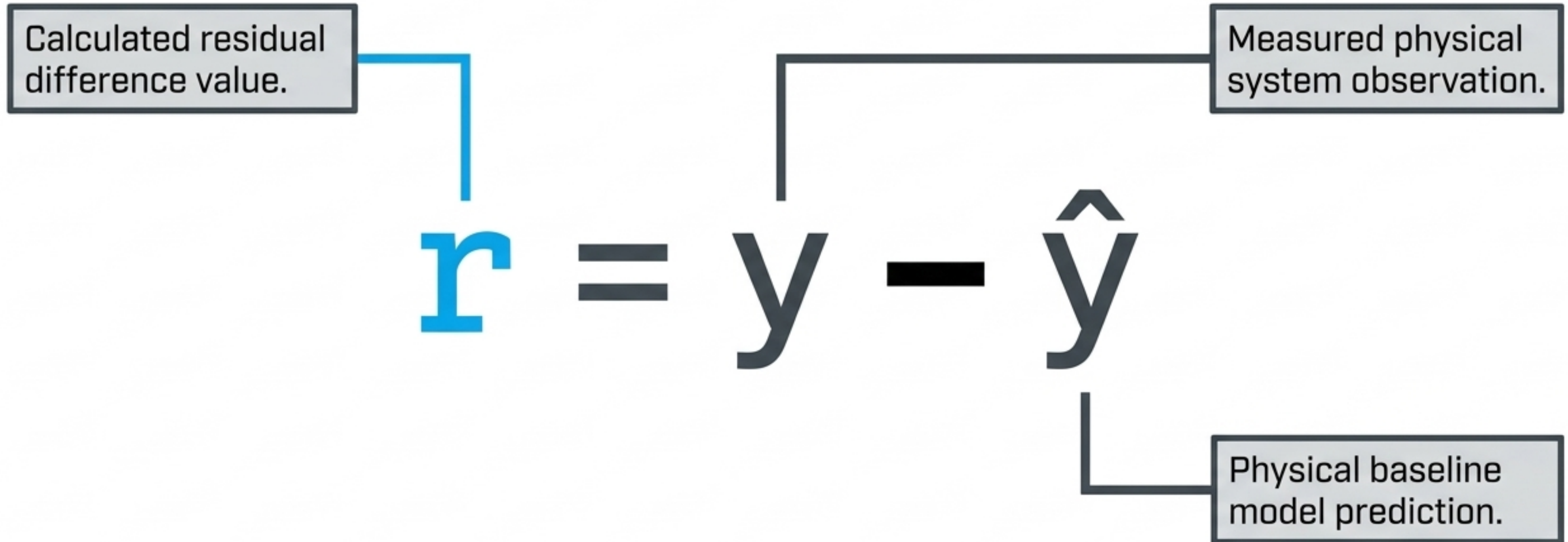
η : Dimensionless overall system efficiency parameter.

Efficiency accounts for mechanical and electrical losses.

3. Hybrid Application

Calculating residuals and
integrating learned corrections.

Defining Model Residuals



Residuals quantify unmodeled physical effects or errors.

Baseline and Residual Calculation

Given parameters: $\Delta p = 100 \text{ kPa}$, $Q = 0.02 \text{ m}^3/\text{s}$, $\eta = 0.8$.



Measured observation y : 2.8 kW .



Hydraulic baseline P_h : $100,000 \text{ Pa} \times 0.02 \text{ m}^3/\text{s} = 2,000 \text{ W}$.



Input prediction $\hat{y}$: $2,000 \text{ W} / 0.8 = 2,500 \text{ W}$.



System residual r : $2.8 \text{ kW} - 2.5 \text{ kW} = +0.3 \text{ kW}$.

Paradigm Synthesis Matrix

Approach	Relationship Source	Application Example
Physics	Equations and assumptions	Pressure rise × flow
Data	Measured observation examples	Fitted electrical-power model
Hybrid	Physics with learned variables	Residual correction integration

Key Points

- Models classify as physical, data-driven, or hybrid.
- Hydraulic power modeling requires strict SI unit consistency.
- Hybrid models apply learned corrections to physical baselines.